# NAPCORE Helpdesk: From Search To Grounded Chat Helpdesk

## Presentation Draft

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# Problem And Goal

- Need a trusted answer layer for multimodal standards implementation.
- Main risk is fast but uncited guidance that creates delivery risk.
- Goal is practical answers with traceable references and controlled escalation.

# System Context (C4 Level 1)

**Five stakeholder groups share one helpdesk interface:**

- PTO (Public Transport Operator) — responsible for daily service delivery
- PTA (Public Transport Authority) — sets policy and compliance standards
- Developer — implements integrations and APIs
- ITS System Integrator — connects helpdesk to broader mobility ecosystem
- Ticketing Agent — manages passenger-facing systems

**External systems:**

- Approved GitHub repositories (NeTEx-CEN/NeTEx) — knowledge source
- LLM provider — grounded generation only; no unconstrained model memory

**Key architectural constraint:**

- All answers sourced from approved repositories only

# Scope

- Standards: Transmodel, NeTEx, SIRI, OJP/OpRa, DATEX II.
- Priority source: NeTEx-CEN/NeTEx and approved companion repositories.
- Demo surfaces are now split into `/user` and `/operator` routes over one shared backend/session shell.

# Evolution Of Helpdesk Implementations

- Stage 1: keyword search over standards documents.
- Stage 2: grounded Q and A with FAQ-first orchestration.
- Stage 3: editorial workflow, routing, and KPI visibility.
- Stage 4: chat-style sessions ready for deterministic and LLM-backed generation.
- The trust model stays fixed: allow-listed repositories, citations, and review gates.

# Why The Evolution Matters

- Search helps lookup, but not decision-ready guidance.
- FAQ-first gives low-latency canonical answers for recurring issues.
- Editorial workflow converts runtime demand into governed knowledge.
- Chat sessions improve usability without weakening provenance.
- Companion repositories expand coverage without relaxing the trust boundary.

# Core Approach

- FAQ-first with RAG fallback.[1]
- FAQ-first: return approved answers when confidence is high.
- RAG fallback: retrieve source chunks, then generate a grounded answer.
- Generation profiles: deterministic-grounded now, LLM-ready when configured.
- If evidence is weak, abstain explicitly.

[1] RAG fallback means that when no high-confidence FAQ answer is available, the system retrieves relevant source chunks and then generates an answer grounded in that retrieved evidence.

# Trust And Safety

- Repository allowlist enforces the knowledge boundary.
- Generation is limited to retrieved evidence.
- Citation gate blocks uncited publication.
- Editorial gate controls draft, review, approve, and publish states.

# Functional Requirements Highlights

- FR-001: question answering through the helpdesk UI.
- FR-004: ingestion and indexing of approved GitHub sources.
- FR-005: retrieval and grounded generation.
- FR-006: anti-hallucination and boundary enforcement.
- FR-002 and FR-003: editorial curation and FAQ promotion.

# Data Model Logic

- Source governance: approved_repositories -> source_documents -> source_chunks.
- FAQ lifecycle: faq_entries -> faq_versions -> review_events.
- Runtime telemetry: question_events -> retrieval_events.
- Evidence mapping: answer_evidence_links ties answers to source chunks.

# User Experience Flow

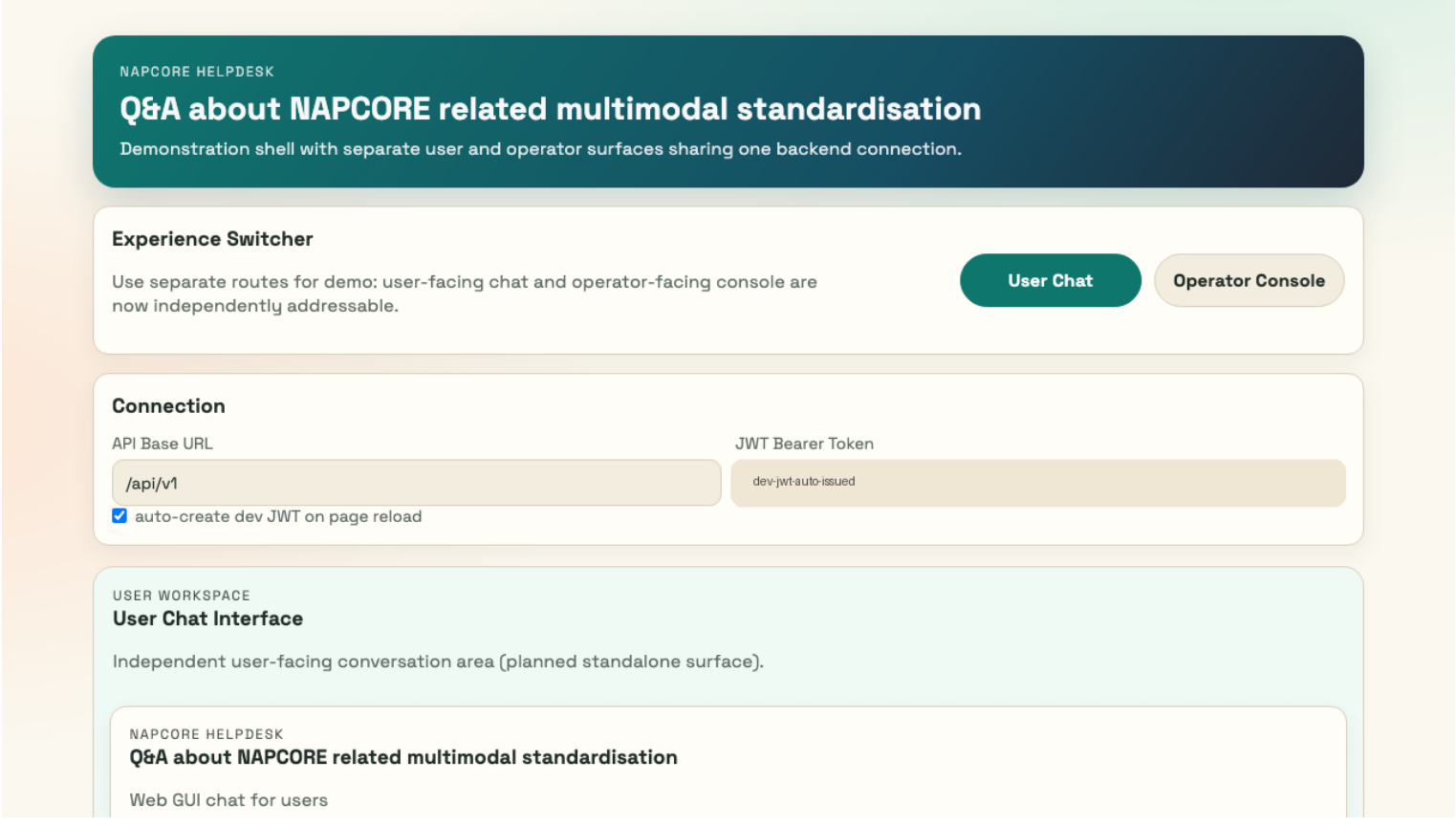
- User opens `/user` for conversation or `/operator` for operational review.
- System attempts FAQ match first.
- On miss or low confidence, the system executes retrieval plus grounded generation.
- Response includes references, confidence, and request trace.
- Session history supports iterative questioning without losing provenance.
- Feedback and editorial routing support continuous improvement.

# Current Product State

- Frontend now exposes separate routed surfaces for user chat and operator console.
- Backend supports deterministic grounded generation and an optional LLM-ready mode.
- Knowledge index is built from approved repositories and selected companion repositories.
- Shared connection state keeps auth and backend targeting consistent across both routes.

# Current Chat UX

- Live `/user` chat route with per-turn evidence, confidence, and request trace.
- Separate `/operator` route handles orchestration, board, KPI, and editorial actions.



# Operations And Governance

- Reviewer decisions are logged and auditable.
- Editorial operations run through Django Admin.
- Recurring intents are promoted into a curated FAQ backlog.
- Re-indexing handles source and standards updates.
- Metrics monitor quality and safety continuously.

# MVP Success Metrics

- Citation coverage for published FAQs: 100%.
- Unsupported-claim publication rate: 0%.
- P95 latency: $<=$ 8 seconds.
- Retrieval success on covered intents: $>=$ 90%.

# Current Artifacts

- Functional requirements, testing pack, and local run guide.
- RAG architecture, database logic, and updated C4 architecture documents.
- Technology baseline: Django 5 + DRF backend, React/Vite frontend, editorial workflow.
- Separate routed user/operator UX, deterministic grounded answering, and LLM-ready generation adapter.
- PlantUML and exported SVG architecture assets for reproducible documentation.

# Next Steps

1. Complete pilot testing across chat UX, editorial workflow, and promotion flows.
2. Enable and evaluate grounded LLM mode against deterministic fallback behavior.
3. Refine ingestion profiles for companion repositories and standards-specific scopes.
4. Use pilot evidence to decide production hardening and deployment model.

# Appendix: Example Questions

- How to use NeTEx for exchanging a timetable?
- How to implement IDs for a Stop Place registry?

Expected outputs: - practical implementation guidance, - explicit citations, - abstention where evidence is insufficient.